Application and Discussion of Airbag in Salvage

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Abstract. The environment of wrecks and sunken objects in the sea is ever-changing, and the salvage methods are varied. Because of its unique performance and advantages, air bag has shown great potential and value in wrecks salvage and water transportation. Air bags have always been used as buoyancy AIDS in the field of salvage, usually tied to the two sides of the wreck or placed in the wreck cabin, and then combined with internal buoyancy or other large machinery to complete salvage. In some complex narrow waters, when large machinery cannot play a role, air bag buoyancy has become the best choice. This paper introduces the development and application scenarios of air bag salvage technology, summarizes the basic principle and technical characteristics of air bag salvage, and summarizes the key technologies of air bag salvage through the research of the construction technology of air bag salvage, and discusses the limitations and prospects of its application.

Keywords: Airbags · Salvage · Construction technology · Shipwreck

1 Introduction

China is a maritime power with a mainland coastline of over 18,000 km, as well as over 5000 rivers and over 500 lakes of various sizes. Its water transportation conditions are well-developed. With the continuous development of the economy, the maritime industry is also becoming increasingly prosperous, with a rapid increase in the number and tonnage of ships, an increase in the density of maritime transportation, and an increase in the range of navigation. Affected by meteorological changes, natural disasters pose a serious threat to the safety of people's lives and property, and major maritime accidents occur frequently. In addition to clearing obstacles in ports and waterways, the salvage of some special sunken targets also falls within the scope of emergency rescue and salvage, such as the salvage of the "5.7" air disaster black box [1], the maritime emergency support mission of spacecraft, and the clearance and salvage of hydropower stations. The main characteristics of such emergency rescue and salvage tasks are difficulty in underwater search or narrow construction waters, and relatively light salvage weight.

After more than half a century of continuous development, the equipment and techniques for salvage are also constantly being updated. The salvage methods can be divided according to the integrity of wrecks and objects into: Overall salvage, segmented salvage, salvage of broken ships, and dismantling salvage. According to the nature of buoyancy,

© The Author(s) 2026
D. Wan et al. (Eds.): JCCME 2024, LNCE 735, pp. 711–723, 2026.
https://doi.org/10.1007/978-981-95-1487-8_70

it can be divided into internal buoyancy salvage method and external buoyancy salvage method [2]. The internal buoyancy salvage method can be divided into: Sealed pumping salvage, compressed air drainage salvage, sealed inflation drainage salvage, and filling material salvage; The external buoyancy salvage method can be divided into: Float bowl salvage method, floating crane salvage method, lifting and prying salvage method, etc. [3]. Airbag salvage can be used for both external and internal buoyancy salvage. The external buoyancy salvage of airbags is similar to the float bowl salvage method, in which the airbags are fixed to the wreck's sunken object using a sling or steel structure frame. Airbag internal buoyancy salvage is a part of the filling material salvage method, which involves inflating the airbag into the cabin to restore the buoyancy inside the ship.

The development process of airbag salvage technology is closely related to the continuous deepening of human exploration and utilization of marine resources. As early as ancient times, people had already used simple buoyancy principles to carry out salvage work using primitive tools such as animal skins filled with air. Entering modern times, with the progress of materials science and the development of industrial technology, rubber airbag salvage technology has undergone a transformation from primitive to modern. From Fig. 1, it can be seen that since the successful trial of rubber airbags for ship launching in 1981, the application of rubber airbags underwater [4] has gradually gained attention. In the 1980s, ship airbag launching technology was promoted in some inland shipyards in China, but due to structural and material strength limitations, the supported tonnage of ships was relatively small, usually with a self-weight of no more than 500 t. With the development of technology, the tonnage of ships that airbags can withstand has gradually increased, and by 2007, they were able to withstand the launch of ships with a self-weight of 9000 t underwater.

Fig. 1. Rubber bag launching

At present, rubber airbags are widely used in ship loading and launching construction due to their advantages of fewer limiting factors, no need for large unloading equipment, short construction period, and low cost. Two industry standards have been formed, CB/T-3795 "Rubber Airbags for Ship Loading and Unloading" and CB/T-3837 "Requirements for Rubber Airbags for Ship Loading and Unloading Process" [4]. In addition, rubber airbags have played an important role in fields such as construction [5], rescue, and pipeline sealing [6]. In addition to rubber airbags, there are currently materials such as polyurethane, PVC, and coated fabrics for airbags, which are widely used in fields such as automobiles and hovercraft [7].

Airbags mainly appear in the field of rescue in the form of buoyancy assistance, which is to cooperate with the distressed ship's own buoyancy or other main lifting equipment to complete the lifting and shallow work of the distressed ship. In the salvage industry, rubber airbags are often placed inside the cabin to restore buoyancy, and there are also some cases where external rubber airbags are used to provide auxiliary buoyancy, and there are few cases where external single rubber airbags are used to salvage wrecks and sunken objects. In conjunction with Fig. 2, in 2017, the Shanghai Salvage Bureau completed the salvage project of the "Sewol" ship and used 3.5-m diameter rubber airbags as buoyancy aids fixed to the ship's structure to assist in the lifting operation of the bow [8]. In conjunction with Fig. 3, in 2020, the China Yantai Salvage used a 4 m diameter external single rubber airbag for salvage operations in the "xxx hydropower station concrete foreign object" salvage project due to factors such as limited operating space and lack of large mechanical equipment. The concrete foreign object was successfully moved to the designated area.

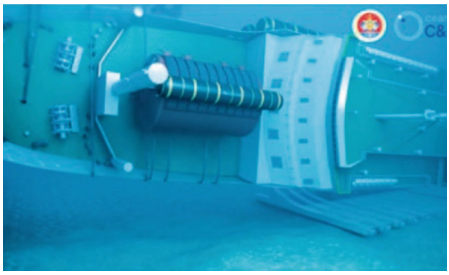


Fig. 2. The Sewol with external auxiliary rubber airbags [9].

Fig. 3. Single external rubber airbag salvaged foreign bodies in hydropower station

2 The Application of Airbags in Salvage Engineering

2.1 Application Case of Airbag External Buoyancy Salvage Method

In March 2020, during an underwater inspection of the sand discharge system at a certain hydropower station, divers discovered varying degrees of defects in the sand discharge tunnel and bottom hole through underwater exploration and inspection. Among them, there was a large concrete foreign object in front of the entrance inspection door of the 5 # sand discharge tunnel. Due to the limited strength of the dam crest of the hydropower station, it is unable to support large lifting equipment. Therefore, the foreign object can only be lifted and displaced by external buoyancy devices such as buoys and rubber airbags. Due to the width of the sand hole being only 4.5 m, the weight of the foreign object reaches nearly 120 t, and the length of the foreign object is only 7.7 m, making it impossible to place multiple rubber airbags or buoys. In order to provide more buoyancy in limited space and facilitate operation, an external single rubber airbag salvage solution was ultimately chosen. At the same time, when using rubber airbags for floating, the airbags are kept stable to ensure that the concrete blocks rise steadily horizontally during underwater floating, and to ensure that the underwater disposal of concrete blocks will not roll and collide with the edge piers on both sides of the gate or cause blockage. Finally, the Yantai Salvage Bureau successfully applied this technology and lifted and displaced foreign objects in the concrete. As shown in Figs. 4 and 5.

Fig. 4 Stabilizing air bag

2.2 Application Case of Buoyancy Salvage Method Inside Airbag

On April 16, 2014, the South Korean passenger ship Shiyue sank in the southwest waters of South Korea, resulting in 304 deaths and 9 missing. Due to the weight of the wreck on the seabed being over 7000 t, the construction unit considered factors such as the

Fig. 5 Rubber airbags help float concrete foreign bodies

structural strength of the hull, and fully utilized the internal and external buoyancy of the wreck during the lifting process of the bow. The internal buoyancy adopts the scheme of compressed air drainage and filling the cabin with airbags. As shown in Figs. 6.

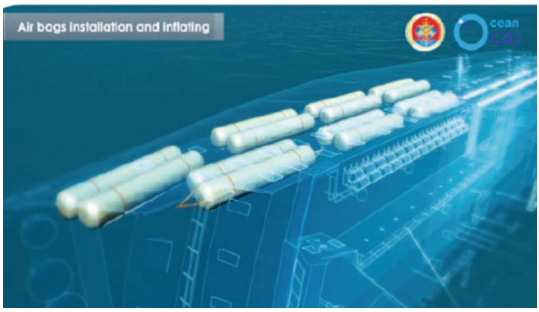


Fig. 6 The wreck filled with air bags [9].

3 Key Technologies for Airbag Salvage

When using rubber airbags in salvage engineering, it is necessary to consider the size, placement, rooting point, strength, operability, and other factors of the rubber airbags.

3.1 Key Points of Airbag Structure Design

The structural design of airbags [10, 11] is the key to ensuring their optimal performance in underwater salvage operations, involving multiple aspects such as material selection, shape design, size calculation [12], pressure resistance [13], and buoyancy distribution. In terms of material selection, the mechanical properties and anti-aging capability required by the airbag must be taken into account. Generally, rubber and polyester fiber canvas

with high strength, high abrasion resistance and high elasticity are used as the main materials. These materials not only need to have good elasticity and tear resistance, but also need to have sufficient resistance to ultraviolet radiation and seawater corrosion. The shape of the airbag used for salvage is usually cylindrical, which can increase the contact area and also be assembled in narrow spaces. The design of pressure resistance is the key to ensuring the safety of airbags. When designing, it is necessary to consider the water pressure effect of airbags at different depths and the impact of gas pressure on the expansion of airbag materials during inflation. At the same time, it is also necessary to consider the floating speed of wrecks to match suitable safety valves and quantities.

The general design process for airbags is:

1. Select the airbag body material according to the requirements, taking into account the material properties such as resistance to bursting strength, plasticity, and tensile strength;
2. Determine the geometric dimensions of the capsule body. The size of airbags, including diameter and length, needs to be determined based on factors such as the space in the wreck compartment and the buoyancy required for airbags.
3. Calculate the working pressure. Calculate the working pressure of the airbag based on factors such as water depth, and the inflation pressure of the airbag cannot exceed the allowable pressure.
4. To determine the number of safety valves, the following method can be used for calculation.

According to the gas pressure formula, assuming that the temperature inside the airbag is conserved during the lifting process, the volume change before and after lifting is:

$$P_1 V_1 = P_2 V_2 \tag{1}$$

Among them, P1 and P2 represent the pressure before and after airbag lifting; V1 and V2 are the volumes before and after airbag lifting, and the volume of exhaust before and after lifting can be calculated using this equation.

According to the Bernoulli equation, there are:

$$P_1 + \frac{\rho_1 v_1^2}{2} + \rho_1 g H_1 = P_2 + \frac{\rho_2 v_2^2}{2} + \rho_2 g H_2 \tag{2}$$

Among them: ρ1 and ρ2 is the density of gas inside the airbag; v1 and v2 are the exhaust velocities of the safety valve, with v1 = 0 in the initial state; H1 and H2 are the water depths before and after lifting.

If the lifting speed of the ship is v, the lifting time is:

$$t = \frac{(H_2 - H_1)}{v} \tag{3}$$

Substitute the above results into the following gas flow rate formula:

$$v = v_2 St \tag{4}$$

The total exhaust hole area S can be obtained, and the number of safety valves can be calculated according to the following equation:

$$S = \frac{N\pi d^2}{4} \tag{5}$$

Among them, d is the aperture of the safety valve, and N is the number of safety valves.

3.2 Rubber Airbag Model Test

At present, the application of airbags in the field of salvage engineering is mainly based on rubber airbags. Therefore, this section takes a rubber airbag with a diameter of 1 m and an effective length of 6 m as an example for corresponding tests, with a working pressure of 0.5 MPa and a safety factor of 3 times.

Airbag Airtightness Test. The airtightness test mainly verifies that the airbag can maintain a certain pressure for a certain period of time, thereby continuously providing buoyancy and reducing the preparation work before the wreck floats. Inflate the airbag to the rated internal pressure, collect data every 12 h, use the pressure correction formula to correct the pressure to the initial temperature, and compare it with the initial pressure. Check whether the airbag has defects such as bubbling and delamination, and record them.

A) Pressure correction formula:

$$P' = \frac{P_2 \times \left(273.15 + T'\right)}{273.15 + T_2} \tag{6}$$

Among them:

P' —— corrected pressure, in megapascals (MPa);

P_2 —— The actual pressure after supplementary inflation of the test airbag, in megapascals (MPa);

T' —— actual measured temperature, in degrees Celsius (°C);

T_2 —— Initial temperature of the test, in degrees Celsius (°C).

B) Calculate the decrease in airbag pressure K according to eq. (2):

$$K = \frac{P - P'}{P \times 100\%} \tag{7}$$

Among them:

K —— Airbag pressure drop rate, (%);

P —— The actual pressure after supplementary inflation of the test airbag, in megapascals (MPa);

P' —— The pressure corrected at a certain time interval, in megapascals (MPa).

From Fig. 7, it can be seen that, the pressure decreased from 0.594 MPa to 0.565 MPa in ten days, a decrease of 0.029 MPa, with a decrease rate of K of 4.88%, which meets the operational requirements.

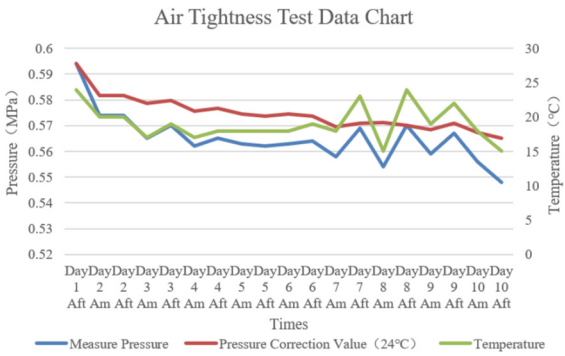


Fig. 7. Air tightness test data

Fig. 8. Air bag bursting test

Blasting Test. The blasting test mainly tests whether the strength of the airbag can reach the design safety **factor**. When the wreck rises rapidly, the external pressure continuously decreases, and the internal gas cannot be discharged in time, whether the airbag is safe. As shown in Figs. 8.

The working pressure of the airbag is 0.5 MPa, and the burst pressure value is 2.18 MPa, which exceeds the design requirement by three times the safety factor. Considering that when a wreck floats, the water pressure decreases and the pressure difference between the inside and outside of the airbag increases, according to the test results, the airbag will not rupture due to excessive pressure during the ascent operation with a water depth difference of no more than 100 m. However, for actual safety reasons, a pressure relief valve needs to be equipped for pressure relief [14].

Based on the experimentally measured burst pressure, the allowable working pressure can be obtained through conversion:

$$p_0 = \frac{D_1 p_e}{nD} \tag{8}$$

Among them:

p_0—— Initial working pressure of airbag (MPa);

p_e—— measured blasting pressure (MPa);

D_1—— diameter of airbag used in blasting test (m);
D—— diameter of working airbag;
N—— Safety factor.

3.3 Safety Control of Airbags

Due to the softness of the airbag itself, when the pressure difference between the inside and outside reaches a certain threshold (the lower limit of the pressure difference between the inside and outside of the airbag), the airbag can provide maximum buoyancy. At this time, the hardness of the outer surface of the airbag is "enhanced" (similar to car tires). As the lifting depth increases, the external pressure will gradually decrease. When the pressure difference between the inside and outside reaches another threshold (the upper limit of the pressure difference between the inside and outside of the airbag), the airbag will tear. In actual salvage projects, it is necessary to choose an appropriate strength airbag based on the water depth. However, due to folding, friction, and other phenomena during transportation, deployment, and other operations, the strength of the airbag will decrease to varying degrees. Therefore, airbags should be equipped with pressure relief valves, and the number of pressure relief valves should be equipped according to the size of the airbags, lifting speed, etc.

3.4 Notes on Airbag Layout Plan

For the internal buoyancy salvage method, priority should be given to filling the airbags into the following types of compartments, including those that cannot restore internal buoyancy and those that are crucial for the buoyancy of wrecks. At the same time, attention should be paid to various factors such as the selection and quantity of airbags, the selection of sinking routes, the portability of inflation, the arrangement of inflation hoses, the selection of airbag fixing points, the verification of force points in the cabin, the anti-puncture treatment of airbags, and the recovery of airbags.

For the external buoyancy salvage method, the focus is on factors such as the placement of airbags, fixation methods, rooting points, stability of airbags, and the coordination between anti puncture treatment and the main lifting method.

3.5 Airbag Fixation and Sinking Control

Due to the irregular shape of rubber airbags before reaching the lower limit of internal and external pressure difference, it is difficult to control their shape during the sinking process, especially when filled in the cabin to provide internal buoyancy, making it difficult for divers to operate underwater.

When using the buoyancy salvage method inside the airbag for salvage operations, it is necessary to select a suitable position to lower the airbag [15], and multiple guides need to be welded in the cabin. Divers use the guides and ship equipment to tow the airbag to the designated position and fix it. Due to the fact that the airbag will float up and be located inside the cabin after inflation, it is necessary to clean the quick opening at the contact position between the airbag and the cabin in advance, and calculate the

ship's strength at the contact position in advance. After sinking, connect the inflatable leather dragon in advance and mark it in order, extending it to the air distribution box.

For the external buoyancy salvage method, nets, steel cages [9], steel pipes, etc. are usually chosen to fix the airbags. As shown in Figs. 9. For small airbags that only require one rooting point, the airbag can be directly connected to the rooting point. External airbags are easier to sink compared to internal airbags. Use a crane to lift the airbag to the designated position, try to lay it flat, and the diver can fix it underwater. Similarly, it is necessary to place the inflatable leather dragon in advance.

3.6 Stability Control of Airbag Inflation

For external rubber airbags with multiple force points, the inflation process is the most unstable state. If not well controlled, the gas will cause one end of the airbag to be high and the other end to be low, resulting in excessive local force. In severe cases, it can cause the rubber airbag to fold and damage, leading to salvage failure. Therefore, before carrying out salvage operations, the external airbag should undergo corresponding inflation stability tests. If there is poor stability during the inflation process, one or more control methods can be combined to control the unstable rubber airbag during inflation, such as encrypted mesh bags, encrypted steel structure cages, and temporary rope fastening.

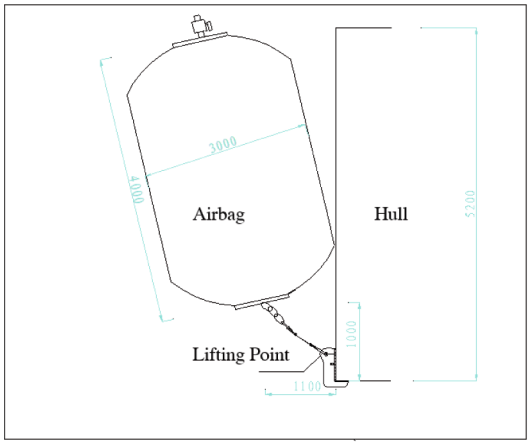
4 Opportunities and Challenges of Airbag Salvage Technology

The commonly used methods for the overall salvage of large wrecks and objects include float bowl salvage, floating crane salvage, etc. Currently, almost all pontoons used in the float bowl method are rigid pontoons made of hard materials. The rigid buoy has a large lifting capacity, but it is easily affected by the underwater environment when diving and being tied to wrecks. In addition, the buoy occupies a large space and has high storage and transportation costs. Large floating cranes are the main tool for salvaging wrecks and objects, but they are often limited by the lifting capacity, operating space, and high transportation costs of cranes. The produced airbag is flexible and versatile, and can be folded or rolled into a cylindrical shape for storage and transportation or diving, greatly improving the salvage ability. When in use, sink the rubber airbag into the seabed, fix it with the salvaged object, and then inflate it into the airbag to float it up, dragging the wreck and object out of the water. Airbags can be inserted into the flooded cabin or fixed to the deck of a wreck, exerting a small force on the unit area of the hull and providing strong support for ship safety. The impact of hydrological conditions on airbag diving is relatively small, and the efficiency of underwater operations is high. Airbags not only provide buoyancy for ship salvage, but also have great advantages in rescuing stranded ships. By means of excavation and other means, airbags can be inserted into the bottom of stranded ships. Inflation of airbags can lift the ship up, and under the towing effect of tugboats, the ship can smoothly resurface [16].

However, due to factors such as material strength and difficulty in lifting control, airbags cannot be scaled up, resulting in smaller drainage volume and lower lifting capacity. At the same time, in order to better control rubber airbags underwater, a method

(a) Steel structure cage fixing method. [9]

(b) Fixing method of net bag sling.



(c) Single root fixation method.

Fig. 9. Air Bag Fixation.

of "breaking them into smaller ones" is usually adopted to divide a large rubber airbag into multiple small rubber airbags, but it also further reduces the buoyancy of the airbag. Therefore, airbags are usually used as auxiliary buoyancy devices, and if they want to be applied on a larger scale in the field of salvage, further issues of material strength and operability need to be addressed.

5 Conclusion

The technology of using airbags to salvage wrecks and objects emerged earlier due to its low cost, low consumption, pollution-free, high efficiency, flexible mobility, and simple operation. However, due to material limitations, it cannot become the mainstream salvage method.

This article introduces the development history of airbag salvage technology for wrecks and sunken objects, and introduces the application scenarios of airbag salvage through two cases: Internal buoyancy salvage and external buoyancy salvage. This article outlines the basic principles and technical characteristics of airbag salvage, studies the construction process of airbag salvage, summarizes the key technologies of airbag salvage for wrecks and sunken objects, and points out the opportunities and challenges of airbag salvage in the future.

With the advancement of technology, airbag materials are constantly being updated and replaced. From natural rubber to butadiene styrene rubber, butyl rubber, chloroprene rubber [17], chlorosulfonated polyethylene, polyvinyl chloride, as well as thermoplastic polyurethane materials [18] and ultra-high molecular weight polyethylene [19], the strength, airtightness, and wear resistance have been greatly improved. Now it is possible to produce airbags with larger drainage volumes, and according to different manufacturing processes, airbags can be divided into multiple compartments to better control their stability. With the continuous improvement of performance, airbags will have a wider range of applications in the field of rescue and other fields in the future.

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